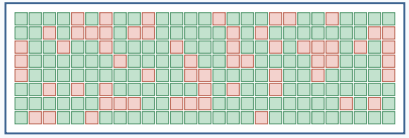
(a) Evaluation data

Production analytics agent
models, prompts and tools evolve

evaluated repeatedly

Benchmark: 519 questions
Each run takes ≈ 3h

574 runs × 519 questions



52 days; pass / fail per cell

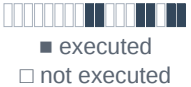
split by time ⟶

287 calibration	**287 held-out**
Day 1 – Day 28	Day 29 – Day 52

+ 299 runs, 5 agent families

(b) Partial-evaluation methods compared

Random sampling
baseline



k questions drawn at random

ŝ = avg(executed verdicts)

Historical caching
reuse stable verdicts



skip questions with recent pass rate ≤ ε or ≥ 1−ε

ŝ = avg(executed + cached)

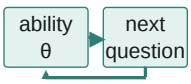
Fixed subset
chosen once from history



representative questions, reused every run

ŝ = Σ wᵢ yᵢ or gp-IRT

Adaptive testing
run-specific selection



most informative question next

ŝ = p-IRT reconstruction

(c) Fidelity metrics

Execution



executed fraction |S| / N

Score fidelity

estimate ŝ

reference s (full run)

MAE = mean | ŝ − s |
over the 287 held-out runs

Ranking fidelity

run order by ŝ by s



Spearman ρ and Kendall τ
between the two orderings